themselves provide management and troubleshooting tools. There are also third-party offerings for Hadoop management from PlatformComputing, which was acquired by IBM *(www.platform.com/Products/MapReduce)* and Zettaset *(www.zettaset.com/)*.

There are even a number of open source alternatives to Hadoop. Some of the popular ones are Disco *(discoproject.org/)*, Spark *(www.spark-project.org/)*, BashReduce *(blog.last.fm/2009/04/06/mapreduce-bash-script)*, GraphLab *(graphlab.org/)*, Storm *(tech.backtype.com/)*, HPCC systems, etc. Disco is an offering from Nokia, which allows developers to write MapReduce jobs in Python, while its back-end is built using Erlang, the functional language, which has built-in support for consistency, fault tolerance and job scheduling. Disco does not use HDFS, but uses a fault-tolerant distributed file system DDFS. Spark was developed by UC Berkeley *(amplab.cs.berkeley.edu/)* and it allows in-memory MR processing distributed across multiple machines. Spark is implemented using Scala.

While I cannot cover the details of each open source Hadoop alternative, I would like to make a mention of HPCC systems from LexisNexis *(hpccsystems.com/)*. HPCC has its own ecosystem equivalent of Hadoop, with its own distributed file system. It supports the development of massively parallel processing jobs by allowing them to be specified in a high-level data-declarative language known as Enterprise Control Language (ECL).

There are also a number of application software products based on Hadoop. These typically simplify writing MapReduce jobs for Hadoop, or enable the analysis of data stored in Hadoop clusters through alternate means rather than the traditional MR paradigm. For instance, Karmasphere *(karmasphere.com/index.php)* provides a Hadoop abstraction layer known as Karmasphere Analyst for Hadoop. Hadapt *(www.hadapt.com/)* combines Hadoop and relational DBMS technology into a single system, making it easy to analyse multi-structured data.

Just as SQL and RDBMS marked the beginning of a new era in data management, this is the beginning of a new era in handling big data, with Hadoop heralding new techniques in processing unstructured data. END

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